MIL-T-21150B(SHIPS)
29 December 1967
SUPERSEDING
MIL-T-21150A(SHIPS)
15 September 1959
(See 6.4)

MILITARY SPECIFICATION

TIE DOWN, CARGO AMMUNITION SHIP, 2000-POUND CAPACITY

1. SCOPE

1.1 This specification covers cargo tie down devices to be furnished as complete assemblies, or as individual components, as specified in the contract or order (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

V-T-295 - Thread, Nylon.
QQ-P-416 - Plating, Cadmium (Electrodeposited).
CCC-C-428 - Cloth, Cotton Duck; Fire, Water, Weather, and Mildew Resistant.
PPP-B-576 - Box, Wood Cleated, Veneer; Paper Overlaid.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes; Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-T-76 - Tape, Pressure-Sensitive Adhesive, Paper, (For Carton Sealing).

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-W-530 - Webbing and Tape, Textile, Cotton, General Purpose, Natural or in Colors.
MIL-E-5272 - Environmental Testing, Aeronautical and Associated Equipment, General Specification for.
MIL-W-5665 - Webbing, Textile, Cotton Warp.
MIL-I-6903 - Ink, Marking (For Parachutes and Other Textile Items).
MIL-A-8625 - Anodic-Coatings, For Aluminum and Aluminum Alloys.
MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Preproduction sample. Prior to production, a preproduction sample shall be submitted for the tests specified in 4.2.

FSC 2090

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3.2 Components. The tie down assembly shall consist of the following:

Item	Description	Requirement
1	Spreader unit	3.4.1
2	Extension unit	3.4.2
3	Adjustment unit	3.4.3
4	Attachment fittings	3.4.4

3.3 Design and construction. The tie down assemblies shall be designed and constructed to lash munitions and cargo items to decks, stanchions, and fittings installed in ammunition ships to secure against movement in a seaway. The assembly shall consist of three cotton webbing straps with appropriate metal fittings. These straps may be used singly, or in combination to suit the load requirement. Means shall be provided for manually adjusting, tensioning, locking, and quickly releasing the assembly. The design of the assembly shall be simple and so constructed that no parts shall work loose in service.

3.3.1 Webbing. Webbing shall be 1-3/4 inch cotton webbing conforming to types VIII and X of MIL-W-5665 and 3 inch hard cotton webbing conforming to type III of MIL-W-530. All webbing shall be treated for mildew resistance, flame retardance, and water repellency. All ends of the webbing shall be protected to prevent fraying. Wearing strips shall be added at all points of wear not already protected by metal rollers. The end of the 1-3/4 inch webbing through the buckle shall have a tab or fold-over to prevent the loss of the strap through the buckle.

3.3.1.1 Stitching. The webbing stitching shall pass the proof load and ultimate load tests specified in 4.5.1.3 and 4.5.1.4. The thread used in stitching the webbing shall be in accordance with type I or II, class 1 of V-T-295.

3.3.2 Hardware. The hardware shall be of a type approved by the command or agency concerned. The hardware shall be free from burrs, sharp edges, fins, pits, cracks, laps, and other surface defects. Edges coming into contact with the webbing shall be well rounded. The hardware shall permit manual operation with complete facility in confined areas and under temperature conditions requiring the use of cold-weathr gloves by operating personnel. Where possible, all hardware shall be aluminum or other nonferrous materials. All aluminum alloy parts shall be anodized in accordance with MIL-A-8625. All steel used shall be cadmium plated in accordance with type I, class 2 of QQ-P-416. All hardware, except that on the spreader unit, shall be insulated from the cargo by a 3 inch cotton pad.

3.3.2.1 Buckles. Buckles shall provide for locking and take-up with provisions for quick release and manual adjustment features. Any length between the adjustment limits shown on figure 1 shall be obtained. The automatic locking device shall not release without being manually disengaged. It shall be possible to position the tensioning buckle on the adjustment unit of figure 1 to permit its location within an approximate 30 inch limit, assuring the most accessible location to personnel during the tensioning operation.

3.3.2.2 Tensioning. After a preliminary adjustment of the webbing length, the adjustment unit shall be capable of being manually tightened, applying a tension in the webbing of not less than 500 pounds with an applied force of not more than 75 pounds. This shall be possible over the full length adjustment range of the assembly. A positive lock which automatically engages upon the completion of a tensioning stroke shall be incorporated in the tensioning mechanism. It shall be possible to tension the webbing while the mechanism is against a flat surface or entirely unsupported.

3.3.2.3 Release. Operation of the lock release mechanism shall disengage the handle so that the load of the webbing will not acutate the handle in reverse and endanger the operator. Subsequent to release of the lock, the force required to release a tension of 500 pounds shall not exceed 15 pounds. Upon release of tension it shall immediately be possible to introduce sufficient slack in the webbing so that the tie down hooks may be manually disengaged from the attachment points.

3.3.3 Hooks. Hooks shall engage holes in portable stanchions, slotted decks, attachment fittings, and other hooks. Each hook shall be capable of swiveling 360 degrees. Hooks shall admit to rapid attachment without the use of tools or special manipulation. All hooks shall have keepers to prevent accidental disengagement. Hooks shall be conformed to fit early deck tracks (see figure 2) and 1/2 inch diameter punched holes in portable stanchions. Throat opening of 1 inch minimum is required. When tie downs are ordered for systems described in 6.1.2, hooks are not required to engage deck tracks.

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3.3.4 Attachment fittings. The attachment fittings (see figure 2) shall engage and lock into deck tracks. A suitable ring or other device capable of 360 degree rotation to suit tie down configuration, shall be provided. The fittings shall be capable of one-hand engagement, locking, and disengagement. Attachment fittings will ordinarily be ordered only for those systems described in 6.1.2.

3.4 Details of components.

3.4.1 Spreader unit. The spreader unit shall provide two hook points (see figure 1) for anchoring the tie down to the deck tracks, attachment fittings in the deck track, securing holes in portable stanchions or to appropriate parts of the ship's structure. The base shall be 1-3/4 inch webbing. A delta-ring attachment point shall be centrally located between the end hooks.

3.4.2 Extension unit. The extension unit shall provide added length to the assembly when used. The base shall be 3 inch webbing. Sewed to the base shall be the 1-3/4 inch webbing with the hardware. The hardware shall consist of a hook at one end, a hook engaging delta-ring at the other end, and two hook engaging delta-rings (pick-up points) between the ends as shown on figure 1. The pick-up points shall be capable of withstanding a strain within a 180 degree arc. The length and arrangement shall be shown on figure 1.

3.4.3 Adjustment unit. The adjustment unit shall be used to tension the assembly. The base shall be 3 inch webbing. Sewed to the base shall be the 1-3/4 inch webbing with the hardware. The hardware shall consist of a hook at each end and a buckle approximately in the center of the unit. The adjustment shall work as a single unit, or in conjunction with either, or both of the other two units specified herein (see 3.4.1 and 3.4.2). The length and arrangement shall be as shown on figure 1.

3.4.4 Attachment fittings. The attachment fitting can be used to attach the tie down straps to the deck tracks. It shall be possible to engage and lock, or remove the fitting with one hand (see figure 2).

3.5 Interchangeability. All parts having the same part number shall be dimensionally and functionally interchangeable.

3.6 Identification of product. Markings containing the following information shall be placed on the base of each unit in 1/4 inch high letters using ink conforming to MIL-I-6903:

Federal stock number
Name of manufacturer
Manufacturer's part number
Date of manufacture
"Capacity 2000 pounds"
US

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction tests. Preproduction tests shall be conducted at the place of manufacture or as designated by the procuring activity and shall consist of the tests specified in 4.5.1.1 and 4.5.1.2.

4.3 Sampling.

4.3.1 Lot. All units of the same type (spreader, adjustment, extenison, or attachment fittings) offered for delivery at one time shall be considered a lot for purpose of sampling and inspection.

4.3.2 Sampling for visual and dimensional examination. From each lot, samples shall be selected at random in accordance with MIL-STD-105 at inspection level II and examined in accordance with 4.4. The acceptable quality level (AQL) shall be 2.5 percent defective.

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4.3.3 Sampling for group A tests. From each lot, samples shall be selected in accordance with inspection level L-8 of the appendix to MIL-STD-105 and tested in accordance with 4.5.1.

4.3.4 Sampling for group B tests. From each lot, samples shall be selected in accordance with inspection level L-4 of the appendix to MIL-STD-105 and tested in accordance with 4.5.2. If any item fails in any test, the lot represented shall be rejected.

4.4 Visual and dimensional examination. Each of the sample items selected in accordance with 4.3.2 shall be examined and measured to verify conformance to all of the requirements of this specification not involving tests. Any item having one or more defects shall be rejected.

4.5 Test procedures.

4.5.1 Group A tests. Samples, selected in accordance with 4.3.3, shall be tested as specified in 4.5.1.1 and 4.5.1.2.

4.5.1.1 Release mechanism. The sample tie downs shall be assembled in a testing machine and the release mechanism shall be locked. Load shall be applied at a rate of separation of the heads of 1 inch per minute (under no load) until the testing machine indicates a load of 2000 pounds. While the tie down is still loaded, the release mechanism shall be capable of being released with a force not greater than 25 pounds.

4.5.1.2 Loading tests. Each sample tie down shall be capable of a preliminary adjustment in 15 seconds after which the tie down shall be tensioned at 500 pounds with a force not greater than 75 pounds.

4.5.1.3 Proof loads.

4.5.1.3.1 Adjustment unit and attachment fittings. The unit subjected to the tests specified in 4.5.1.2 shall be attached to the attachment fittings and the assembly placed in a testing machine. The buckle shall be actuated to lock the webbing at approximately 3 feet in length. The load shall be applied at a rate of separation of 1 inch per minute (under no load) until the testing machine indicates a load of 2000 pounds. After 30 seconds time, the load shall be released and the unit and fittings examined. There shall be no failure or damage to the webbing or sewing, or any permanent deformation of the metal parts. There shall be no slippage of the webbing through the buckle.

4.5.1.3.2 Extension unit. The extension unit shall be placed in the testing machine with the hook end fixed. The proof load as specified in 4.5.1.3.1 shall then be applied. The load shall be applied by attaching the moving head of the testing machine in turn to each of the delta-ring pick-up points. After testing the two parts of the strap to 2000 pounds for a period of 30 seconds the unit shall be released and examined. There shall be no failure or damage to the webbing or sewing, or any permanent deformation of the metal parts.

4.5.1.3.3 Spreader unit. The spreader unit shall be assembled in the testing unit with the hook ends fixed and approximately 12 inches between them. The moving ram of the testing unit shall be attached to the delta-ring in the center of the strap and the proof load as specified in 4.5.1.3.1 shall be applied. After testing the strap to 2000 pounds, the load shall be released and the unit examined. There shall be no failure or damage to the webbing or sewing, or any permanent deformation of the metal parts.

4.5.1.4 Ultimate load.

4.5.1.4.1 Adjustment unit and attachment fittings. The unit subjected to the tests in 4.5.1.2 and 4.5.1.3.1 shall be installed in the testing machine with the attachment fittings. The buckle shall be actuated to lock the webbing at 3 feet in length. Load shall be applied at a rate of separation of the heads of the machine of 1 inch per minute (under no-load) until the testing machine indicates a load of 3500 pounds. After 10 seconds the load shall be released and the assembly examined. There shall be no failure of the webbing, sewing, or metal parts. There shall be no slippage of the webbing through the adjustment device. Damage such as (a) broken sewing thread without complete separation; (b) broken or frayed webbing threads without complete separation of the webbing; and (c) deformation but not rupture of the hardware, will be permitted. The release mechanism of the buckle shall be manually operable after the test.

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4.5.1.4.2 Extenison unit. The extension unit shall be assembled in the testing machine with the hook end fixed. The ultimate load as specified in 4.5.1.4.1 shall than be applied. The load shall be applied by attaching the moving ram of the testing machine in turn to each of the two delta-rings at the pick-up points. After subjecting each of the two parts of the strap to 3500 pounds for 10 seconds, the unit shall be examined as specified in 4.5.1.4.1.

4.5.1.4.3 Spread unit. The spreader unit shall be assembled in the testing machine with the hook ends fixed and separated by about 18 inches. The moving ram of the testing machine shall be fixed to the delta-ring in the center of the strap and the ultimate load and inspection as specified in 4.5.1.4.1 applied.

4.5.2 Group B tests. Samples, selected in accordance with 4.3.4, shall be tested as specified in 4.5.2.1 through 4.5.2.4, inclusive.

4.5.2.1 Salt spray. One sample tie down shall be subjected to the salt spray tests of MIL-E-5272. The duration of the test shall be 50 hours. There shall be no evidence of corrosion that might affect operation of the tie down.

4.5.2.2 Vibration. One sample tie down shall be subjected to the vibration tests, procedure I, of MIL-E-5272. The tie down shall be kept under a tension of 200 pounds for a period of 24 hours preceding the test. The tie down shall be vibrated under a tension of 200 pounds. No appreciable loss of tension shall result from slipping of the webbing.

4.5.2.3 Low temperature. The tie down sample which has undergone the test specified in 4.5.2.2, shall be subjected to the low temperature tests, procedure II, of MIL-E-5272. While at the specified temperature the adjustment, locking, and release mechanism shall be tested for ease of operation.

4.5.2.4 Flame retardance. The tie down sample shall meet requirements for fire resistance tests as specified in CCC-C-428.

5. PREPARATION FOR DELIVERY

(Preparation for delivery requirements specified herein apply only to direct Government procurement. Preparation for delivery requirements between contractors and sub-contractors shall be as specified in the individual order.)

5.1 Preservation and packaging (see 6.2).

5.1.1 Level A. Tie downs shall be individually preserved and packaged in accordance with method III of MIL-P-116. Contact preservative is not required.

5.1.2 Level C. Tie down shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 Packing (see 6.2).

5.2.1 Level A. Tie downs, preserved and packaged as specified, shall be packed in overseas type wood cleated fiberboard, nailed wood, wirebound wood, fiber, wood cleated veneer paper overlaid, or wood cleated plywood boxes conforming to PPP-B-591, PPP-B-621, PPP-B-585, PPP-B-636, PPP-B-576 or PPP-B-601, respectively, at the option of the contractor. Shipping containers shall have case liners conforming to MIL-L-10547 and shall be closed and sealed in accordance with the appendix thereto. Case liners for boxes conforming to PPP-B-636 may be omitted provided all joints and corners of the boxes are sealed with minimum 1-1/2 inch wide tape conforming to PPP-T-76. Boxes shall be closed and strapped in accordance with the applicable box specification or appendix thereto. The gross weight of wood or wood-cleated boxes shall not exceed 200 pounds; fiber boxes shall not exceed the weight limitations of the applicable box specification.

5.2.2 Level B. Tie downs, preserved and packaged as specified, shall be packed in domestic type wood cleated fiberboard, nailed wood, wirebound, cleated plywood, or wood cleated veneer paper overlaid boxes, or class 2 fiber boxes conforming to PPP-B-591, PPP-B-621, PPP-B-585, PPP-B-601, PPP-B-576, or PPP-B-636, respectively, at the option of the contractor. Box closures shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood or wood-cleated boxes shall not exceed 200 pounds; fiber boxes shall not exceed the weight limitations of the applicable box specification.

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5.2.3 Level C. Tie downs, preserved and packaged as specified, shall be packed in containers which will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply to the Uniform Freight Classification Rules or other regulations as applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, or herein, interior and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The tie down covered by this specification is intended primarily for use in lashing down cargo munitions on Navy ammunition ships incorporating the portable metal dunnage system.

6.1.1 Early metal dunnage systems, as found on AE 21 Class ships have deck tracks configured as shown in figure 2. Attachment fittings on these systems are optional and illustrated in figure 2.

6.1.2 Metal dunnage systems installed in ships since 1966 such as the AOR 1 Class and AE 26 Class have deck tracks with 1-3/8 inch diameter round holes as shown in figure 2. Attachment fittings must be supplied for these systems and shall accommodate the round hole, the track thickness indicated and the clearance below the track as shown.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Applicable level of shipment (see 5.1 and 5.2).
(c) Type of system intended for use with (see 6.1.1 and 6.1.2).

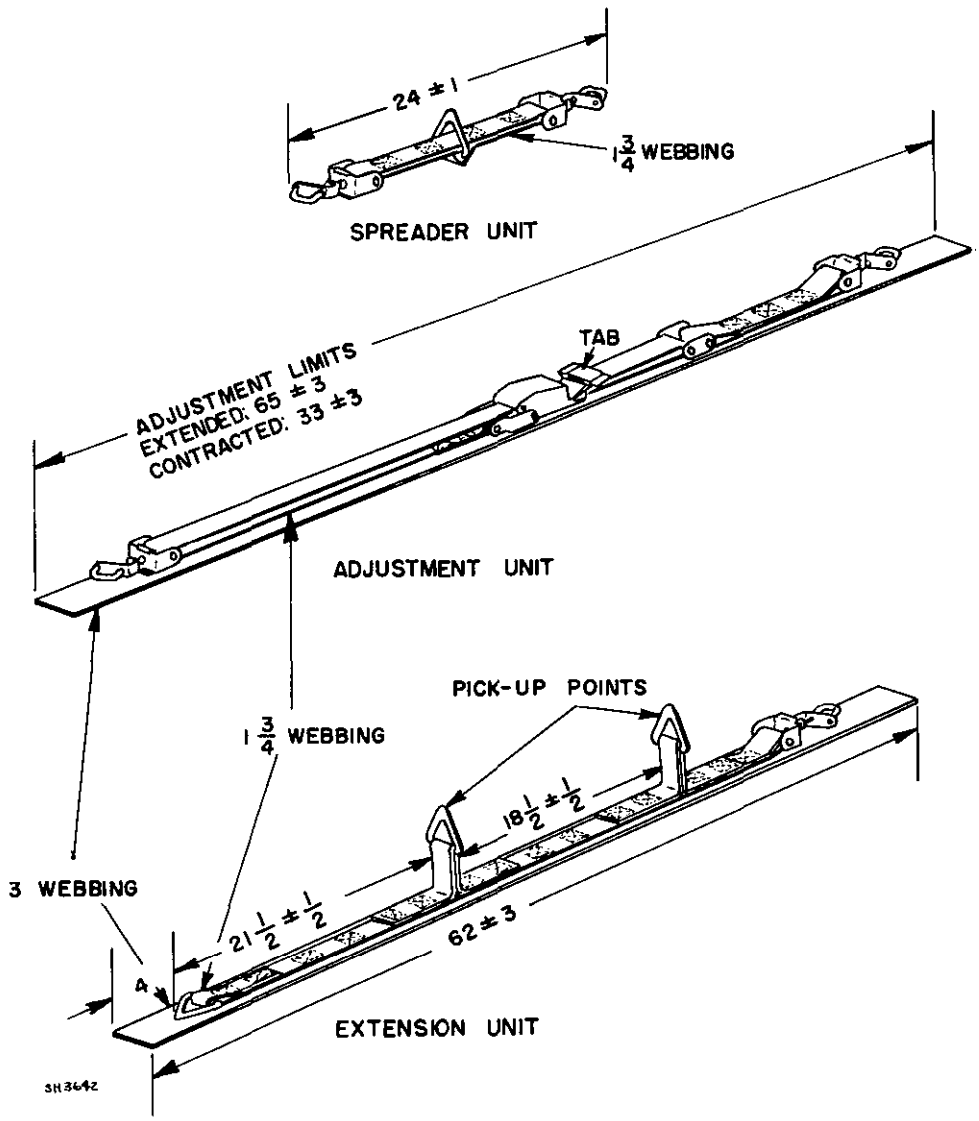
6.3 Preproduction.

6.3.1 Invitation for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.4 *CHANGES FROM PREVIOUS ISSUE.* THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

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NOTE: THIS IS NOT INTENDED TO DELINEATE THE PRECISE DESIGN OF THE UNITS, EXCEPT AS PARTICULARLY SPECIFIED. FOR EXAMPLE, THE DESIGN OF THE BUCKLE IS NOT SPECIFIED BUT THE LENGTH OF EACH UNIT IS SPECIFIED WITHIN LIMITS.

Figure 1 - Envelope sizes and general arrangement for the three units in the assembly.

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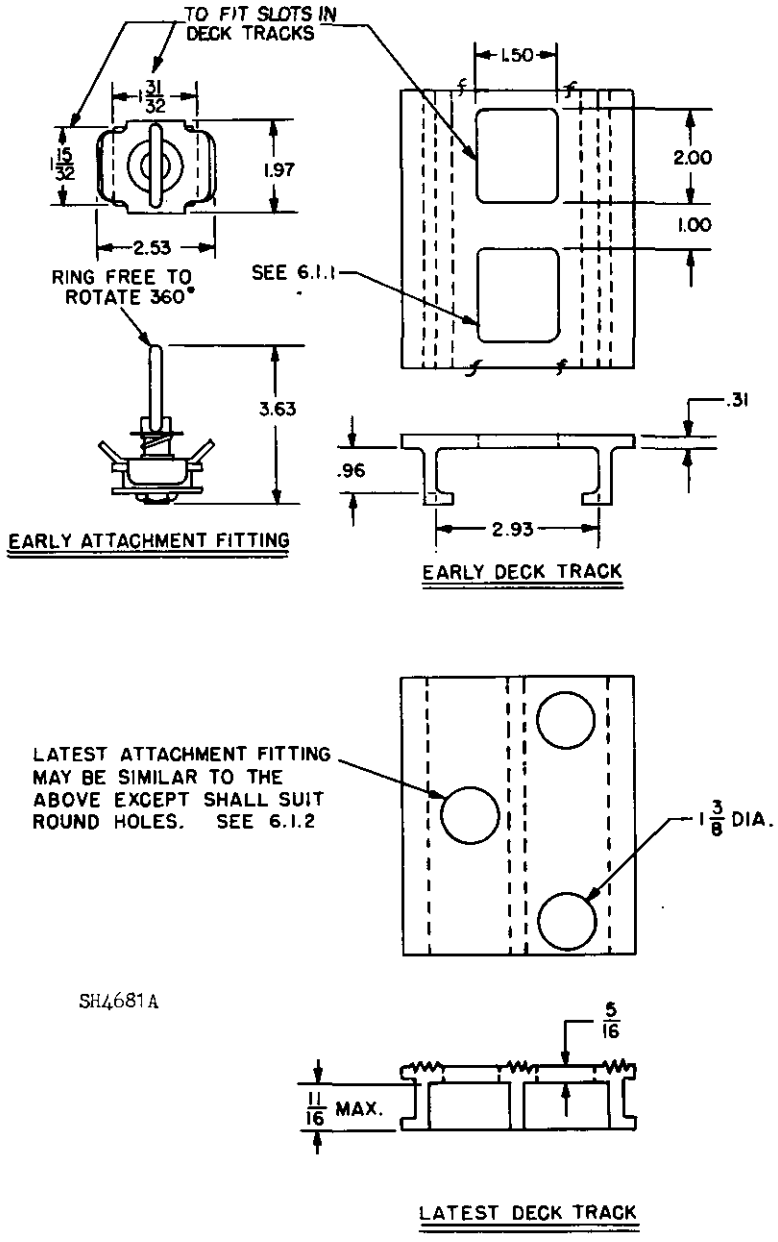


FIGURE 2 – ATTACHMENT FITTINGS AND DECK TRACKS

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